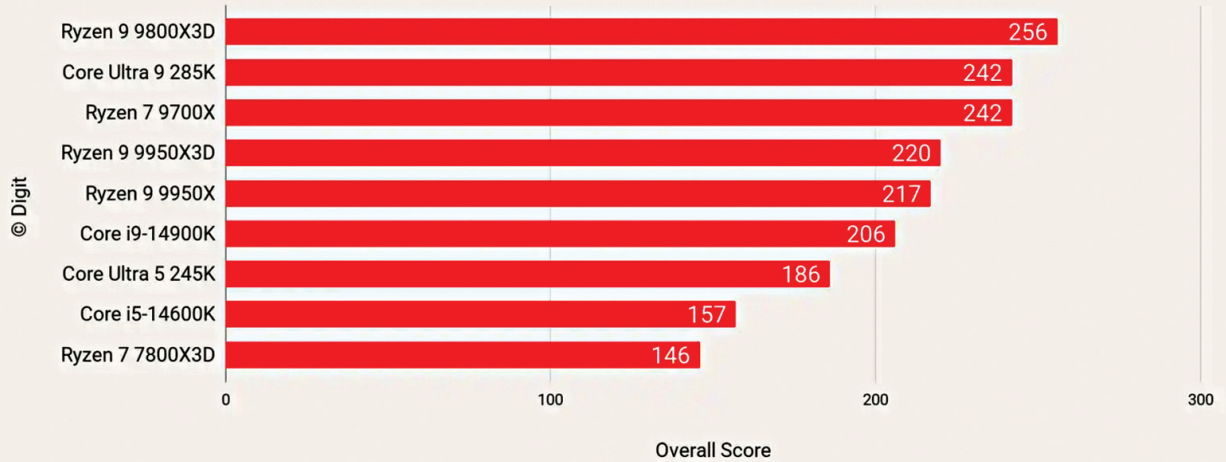


Procyon AI

Computer Vision benchmark



with improving the yield of the chips and helping manufacturers come up with more variations. However, the physical distance between two separate CCDs (Core Complex Dies) also has a latency penalty when the workloads are shared across CCDs. Aside from synthetic benchmarks, its greatest impact is mostly seen in gaming workloads. This is why the higher core-count X3D processors lag behind their lower core-count siblings. That being said, AMD has improved the performance in such scenarios and the 9950X3D has excellent gaming performance. This time around, the performance difference between the 9950X3D and the lower core-count 9800X3D is much lesser than what was observed in previous processor generations. We saw the 9950X3D doing better than the 7800X3D and the 5800X3D. Again, the 9950X3D knocks it out of the park yet again.

CORE TO CORE LATENCY

Core-to-Core Latency measures the time it takes for data to transfer between different cores within a CPU. It highlights the efficiency of the processor's inter-core communication pathways and cache coherency mechanisms.

An interesting insight is that lower core-to-core latency can improve performance in multi-threaded applications where threads need to synchronize or share data frequently. It can reveal architectural differences between CPUs, such as the impact of chiplet designs versus monolithic die layouts, affecting inter-core communication efficiency.

The AMD Ryzen 9 9950X3D has 16 Cores with two CCDs and we can see that within the CCD, the latency for core-to-core communication is around 21-24 milliseconds. And when switching to cores on the other CCD, the latency climbs into the range of 80-83 milliseconds. This is a marked improvement over the older gen processors wherein the communication between CCDs would incur a larger penalty.

THERMALS AND POWER

Being a 16-Core processor does mean that the temperatures tend to skyrocket pretty fast, especially in multi-threaded workloads. With Cinebench and Blender, we did see the Ryzen 9 9950X3D creep up to 84 Degrees pretty fast. Using an AIO becomes absolutely essential with the 9950X3D.

During our throttling test, we saw power draws hitting 200 Watts but

the processor did not throttle. This is a little over the TDP for which the processor is rated for but a good cooler will not only help keep the processor cool but it will also help the inbuilt OC profiles do their job better.

VERDICT

The AMD Ryzen 9 9950X3D is a great processor will all the bells and whistles that one could hope for. The only "bell" missing would be that of power consumption. It does great in single-threaded workloads, content-creation workloads, gaming, ... you name it. And to make all of these improvements worth the money, it should ideally list for a price of INR 70,000 in India. With the older gen 7950X3D still selling for about 65K while consuming much less power, folks who need the power have plenty of options to pick from.

–Mithun Mohandas

SPECIFICATIONS

Cores: 16 | Threads: 32 | Base clock: 4.3 GHz | Boost clock: 5.3 GHz | Process node: TSMC N4 | L3 Cache: 128 MB | L2 Cache: 1 MB per core | TDP: 170 W | Socket: AM5 | Architecture: Granite Ridge

CONTACT

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